

Crash insurance and the G10 carry premium: theory notes

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These notes prove the results listed in `theory/README.md`. Results that are standard are marked as such and attributed; the proofs written here are my own unless a source is named. Throughout, Φ and φ are the standard normal distribution and density functions, $s = \sigma\sqrt{\tau}$, and for forward F and strike K

$$d_{\pm}(K) = \frac{\ln(F/K) \pm s^2/2}{s}.$$

The identity $K\varphi(d_-) = F\varphi(d_+)$ is used repeatedly.

R1. Domestic and foreign risk-neutral measures

This is the change of numeraire of Geman, El Karoui and Rochet (1995); the statement and proof are included for completeness. Girsanov's theorem and Itô's formula are used in the form given by Shreve (2004).

Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \in [0, T]}, \mathbb{Q}^d)$ satisfy the usual conditions, let B^d, B^f be strictly positive adapted money-market accounts with $B_0^d = B_0^f = 1$, and let $S > 0$ be adapted and càdlàg. Assume $Z_t = S_t B_t^f / (S_0 B_t^d)$ is a true $\mathbb{Q}^d$ -martingale and define $d\mathbb{Q}^f / d\mathbb{Q}^d = Z_T$ on $\mathcal{F}_T$.

Proposition 1. (a) $\mathbb{Q}^f$ is a probability measure equivalent to $\mathbb{Q}^d$. (b) For $\mathcal{F}_T$ -measurable $X \geq 0$ or $X \in L^1(\mathbb{Q}^f)$, $\mathbb{E}^{\mathbb{Q}^f}[X | \mathcal{F}_t] = \mathbb{E}^{\mathbb{Q}^d}[Z_T X | \mathcal{F}_t] / Z_t$. (c) An adapted process Y with $Y_t \in L^1(\mathbb{Q}^f)$ is a $\mathbb{Q}^f$ -martingale if and only if ZY is a $\mathbb{Q}^d$ -martingale; in particular a foreign price V^f satisfies: V^f / B^f is a $\mathbb{Q}^f$ -martingale if and only if $SV^f / B^d = S_0 Z (V^f / B^f)$ is a $\mathbb{Q}^d$ -martingale. (d) If $dS_t / S_t = (r_d - r_f) dt + \sigma dW_t$ under $\mathbb{Q}^d$ with constant coefficients, then $W_t^f = W_t - \sigma t$ is a $\mathbb{Q}^f$ -Brownian motion, $d(1/S_t) = (1/S_t)[(r_f - r_d) dt - \sigma dW_t^f]$, and $\mathbb{E}^{\mathbb{Q}^d} S_T = F$, $\mathbb{E}^{\mathbb{Q}^f}(1/S_T) = 1/F$ with $F = S_0 e^{(r_d - r_f)T}$.

Proof. (a) $Z_T > 0$ and $\mathbb{E}^{\mathbb{Q}^d} Z_T = Z_0 = 1$, so $\mathbb{Q}^f$ is a probability measure; positivity of Z_T gives equivalence. (b) For $\mathcal{F}_t$ -measurable $Y \geq 0$, the tower property and the martingale property give $\mathbb{E}^{\mathbb{Q}^f} Y = \mathbb{E}^{\mathbb{Q}^d}[Z_T Y] = \mathbb{E}^{\mathbb{Q}^d}[Z_t Y]$. For $A \in \mathcal{F}_t$,

$$\mathbb{E}^{\mathbb{Q}^f}[X \mathbf{1}_A] = \mathbb{E}^{\mathbb{Q}^d}[\mathbb{E}^{\mathbb{Q}^d}[Z_T X | \mathcal{F}_t] \mathbf{1}_A] = \mathbb{E}^{\mathbb{Q}^f}\left[\frac{\mathbb{E}^{\mathbb{Q}^d}[Z_T X | \mathcal{F}_t]}{Z_t} \mathbf{1}_A\right],$$

which characterises the conditional expectation. (c) Apply (b) on $[s, t]$: $\mathbb{E}^{\mathbb{Q}^f}[Y_t | \mathcal{F}_s] = \mathbb{E}^{\mathbb{Q}^d}[Z_t Y_t | \mathcal{F}_s] / Z_s$, so Y is a $\mathbb{Q}^f$ -martingale exactly when $\mathbb{E}^{\mathbb{Q}^d}[Z_t Y_t | \mathcal{F}_s] = Z_s Y_s$. (d) Here $Z_t = \exp(\sigma W_t - \sigma^2 t / 2)$, so Girsanov's theorem gives the first claim, and Itô's formula for $1/S$ gives the dynamics. The expectations follow from the lognormal law under each measure. $\square$

Remark 2. Jensen's inequality gives $\mathbb{E}^{\mathbb{Q}^d}(1/S_T) > 1/\mathbb{E}^{\mathbb{Q}^d} S_T = 1/F$. There is no contradiction with (d): the foreign investor's forward price is an expectation under $\mathbb{Q}^f$, not $\mathbb{Q}^d$ (Siegel's paradox). If Z is only a strict local martingale, then $\mathbb{E} Z_T < 1$ and Z_T does not define an equivalent probability measure. Carr, Fisher and Ruf (2014) construct the foreign measure in that case as a Föllmer measure that is not equivalent to $\mathbb{Q}^d$, and restore the domestic-foreign symmetry with a modified pricing operator.

R2. Arbitrage-free call prices

The link between the second strike derivative of call prices and state prices is due to Breeden and Litzenberger (1978). I state and prove the characterisation below, with a measure-theoretic proof.

Proposition 3. *Let $c : [0, \infty) \rightarrow \mathbb{R}$ and $F > 0$. There is a random variable $X \geq 0$ with $\mathbb{E}X = F$ and $c(K) = \mathbb{E}(X - K)^+$ for all $K \geq 0$ if and only if (i) c is convex, (ii) $c(0) = F$, (iii) $c'_+(0) \geq -1$ and (iv) $c(K) \rightarrow 0$ as $K \rightarrow \infty$. The law of X is then unique, with distribution function $G = 1 + c'_+$ on $[0, \infty)$ and $\mathbb{P}(X = 0) = 1 + c'_+(0)$.*

Proof. Necessity. Convexity of c follows from convexity of $K \mapsto (x - K)^+$ and linearity of expectation, and $c(0) = \mathbb{E}X = F$. Since $0 \leq (X - K)^+ \leq X \in L^1$ and $(X - K)^+ \rightarrow 0$ pointwise, dominated convergence gives (iv). For $h > 0$, the difference quotient $[(X - K - h)^+ - (X - K)^+]/h$ is bounded by 1 in absolute value and converges to $-\mathbf{1}\{X > K\}$ as $h \downarrow 0$, so $c'_+(K) = -\mathbb{P}(X > K)$. In particular $c'_+(0) = -\mathbb{P}(X > 0) \geq -1$.

Sufficiency. A convex function on $[0, \infty)$ has a right derivative c'_+ that is nondecreasing and right-continuous, and c is absolutely continuous on compact intervals with $c(b) - c(a) = \int_a^b c'_+(u) du$. If $c'_+(K_0) > 0$ for some K_0 , then $c(K) \geq c(K_0) + c'_+(K_0)(K - K_0) \rightarrow \infty$, contradicting (iv); so $c'_+ \leq 0$, c is nonincreasing and, by (iv), $c \geq 0$. Let $L = \lim_{K \rightarrow \infty} c'_+(K) \leq 0$. Since c'_+ is nondecreasing, $c'_+ \leq L$ everywhere, so $c(K) \leq F + LK$; $L < 0$ would force $c < 0$ for large K . Hence $L = 0$.

Define $G(K) = 0$ for $K < 0$ and $G(K) = 1 + c'_+(K)$ for $K \geq 0$. Then G is nondecreasing and right-continuous, $G(0) = 1 + c'_+(0) \geq 0$ by (iii), and $G(K) \rightarrow 1$. It is therefore the distribution function of a random variable $X \geq 0$ (Lebesgue–Stieltjes construction). By Tonelli’s theorem,

$$\mathbb{E}(X - K)^+ = \int_K^\infty \mathbb{P}(X > u) du = \int_K^\infty (1 - G(u)) du = - \int_K^\infty c'_+(u) du = c(K) - \lim_{b \rightarrow \infty} c(b) = c(K),$$

and $\mathbb{E}X = c(0) = F$. Uniqueness holds because c determines c'_+ and hence G . $\square$

Remark 4. Extending c by $c(K) = F - K$ for $K < 0$, the second distributional derivative of c on $\mathbb{R}$ is the law of X ; the jump of c' at 0 carries the atom $\mathbb{P}(X = 0)$. On a finite strike grid the necessary conditions reduce to slopes in $[-1, 0]$ that are nondecreasing, which is the discrete test used in `src/qef/fx/arbitrage.py`. The implementation also applies the density condition on implied total variance of Gatheral and Jacquier (2014, eq. (2.1) and Lemma 2.2).

R3. Spanning

The representation of a payoff by bonds, forwards and out-of-the-money options is the spanning formula of Carr and Madan (2001; first published 1998). The version below states explicit sufficient hypotheses, including the integrability condition under which expectations can be taken term by term; the proof is mine. In R3 and R4, X is the terminal exchange rate and $\mathbb{E}$ and $\mathbb{Q}$ refer to the same pricing measure.

Proposition 5. *Let $f : (0, \infty) \rightarrow \mathbb{R}$ have a locally absolutely continuous derivative f' , so that f'' exists almost everywhere and is locally integrable. Let $X > 0$ almost surely with $\mathbb{E}X = F < \infty$, and write $P(K) = \mathbb{E}(K - X)^+$ and $C(K) = \mathbb{E}(X - K)^+$. If*

$$I = \int_0^F |f''(K)| P(K) dK + \int_F^\infty |f''(K)| C(K) dK < \infty,$$

then $f(X) \in L^1$ and

$$\mathbb{E}f(X) = f(F) + \int_0^F f''(K) P(K) dK + \int_F^\infty f''(K) C(K) dK.$$

Proof. Fix $x > 0$. Absolute continuity of f and f' on the interval between x and F gives $f(x) - f(F) = \int_F^x f'(u) du$ and $f'(u) = f'(F) + \int_F^u f''(K) dK$. Substituting and applying Fubini's theorem on the bounded interval (where f'' is integrable),

$$f(x) = f(F) + f'(F)(x - F) + \int_F^x f''(K)(x - K) dK.$$

For $x > F$ the last integral equals $\int_F^\infty f''(K)(x - K)^+ dK$; for $x < F$ it equals $\int_x^F f''(K)(K - x) dK = \int_0^F f''(K)(K - x)^+ dK$. Hence, for every $x > 0$,

$$f(x) = f(F) + f'(F)(x - F) + \int_0^F f''(K)(K - x)^+ dK + \int_F^\infty f''(K)(x - K)^+ dK.$$

The map $(K, \omega) \mapsto f''(K)(K - X(\omega))^+$ is jointly measurable. By Tonelli's theorem, $\mathbb{E} \int_0^F |f''(K)|(K - X)^+ dK = \int_0^F |f''(K)|P(K) dK$, and similarly for the call term, so both are at most $I < \infty$. Each term on the right is therefore integrable, hence so is $f(X)$, and Fubini's theorem allows expectation and integral to be exchanged. Since $\mathbb{E}(X - F) = 0$, the formula follows. $\square$

Remark 6. For $f(x) = \ln(x/F)$, $f''(K) = -K^{-2}$, which gives the log contract $\mathbb{E} \ln(X/F) = -\int_0^F P(K)K^{-2} dK - \int_F^\infty C(K)K^{-2} dK$. For $f(x) = (\ln(x/F))^2$, $f''(K) = 2(1 + \ln(F/K))/K^2$, which changes sign at $K = eF$; the absolute-value condition in the proposition covers such weights. The moment contracts of Bakshi, Kapadia and Madan (2003, Theorem 1) are of this type; they expand around the spot price with a constant interest rate, whereas the contracts here are centred at the forward F . Quotes observe P and C only between the 10Δ strikes, so the tails of these integrals must be bounded (R4).

R4. Bounds on the unobserved tails

Let $G(u) = \mathbb{Q}(X \leq u)$ and $\bar{G}(u) = \mathbb{Q}(X > u)$. By Tonelli's theorem, $P(K) = \int_0^K G(u) du$ and $C(K) = \int_K^\infty \bar{G}(u) du$. Let $K_{\min} < F < K_{\max}$ be the lowest and highest strikes at which prices are observed.

Lemma 7. (a) If $u \mapsto G(u)/u^\gamma$ is nondecreasing on $(0, K_{\min}]$ for some $\gamma > 0$, then $P(K) \leq P(K_{\min})(K/K_{\min})^{\gamma+1}$ for $0 < K \leq K_{\min}$. (b) If $u \mapsto u^\eta \bar{G}(u)$ is nonincreasing on $[K_{\max}, \infty)$ for some $\eta > 1$, then $C(K) \leq C(K_{\max})(K/K_{\max})^{1-\eta}$ for $K \geq K_{\max}$.

Proof. (a) With $\lambda = K/K_{\min} \in (0, 1]$ and the substitution $u = \lambda v$, $P(K) = \lambda \int_0^{K_{\min}} G(\lambda v) dv$. Since $\lambda v \leq v$ and $G(w)/w^\gamma$ is nondecreasing, $G(\lambda v) \leq \lambda^\gamma G(v)$. Hence $P(K) \leq \lambda^{\gamma+1} \int_0^{K_{\min}} G(v) dv = \lambda^{\gamma+1} P(K_{\min})$. (b) With $\lambda = K/K_{\max} \geq 1$ and $u = \lambda v$, $C(K) = \lambda \int_{K_{\max}}^\infty \bar{G}(\lambda v) dv$, and $\bar{G}(\lambda v) \leq \lambda^{-\eta} \bar{G}(v)$ because $w^\eta \bar{G}(w)$ is nonincreasing. Hence $C(K) \leq \lambda^{1-\eta} C(K_{\max})$. $\square$

Proposition 8. Under the hypotheses of the lemma, for any measurable $w \geq 0$ on $(0, K_{\min}]$,

$$\int_0^{K_{\min}} w(K)P(K) dK \leq \frac{P(K_{\min})}{K_{\min}^{\gamma+1}} \int_0^{K_{\min}} w(K)K^{\gamma+1} dK.$$

In particular $\int_0^{K_{\min}} P(K)K^{-2} dK \leq P(K_{\min})/(\gamma K_{\min})$. For $w(K) = 2(1 + \ln(F/K))K^{-2}$,

$$\int_0^{K_{\min}} w(K)P(K) dK \leq \frac{2P(K_{\min})}{K_{\min}} \left[\frac{1 + \ln(F/K_{\min})}{\gamma} + \frac{1}{\gamma^2} \right].$$

For the upper tail and any measurable w on $[K_{\max}, \infty)$,

$$\int_{K_{\max}}^\infty |w(K)| C(K) dK \leq C(K_{\max})K_{\max}^{\eta-1} \int_{K_{\max}}^\infty |w(K)| K^{1-\eta} dK;$$

in particular $\int_{K_{\max}}^{\infty} C(K)K^{-2}dK \leq C(K_{\max})/(\eta K_{\max})$, and for $w(K) = 2(1 - \ln(K/F))K^{-2}$ with $F < K_{\max} \leq eF$ and $\ell = \ln(K_{\max}/F)$,

$$\int_{K_{\max}}^{\infty} |w(K)| C(K) dK \leq \frac{2C(K_{\max})}{K_{\max}} \left[\frac{1-\ell}{\eta} - \frac{1}{\eta^2} + \frac{2e^{-\eta(1-\ell)}}{\eta^2} \right].$$

Proof. The first inequality is the lemma multiplied by $w \geq 0$ and integrated. For $w(K) = K^{-2}$, $\int_0^{K_{\min}} K^{\gamma-1} dK = K_{\min}^{\gamma}/\gamma$. For the logarithmic weight, integration by parts gives $\int_0^{K_{\min}} K^{\gamma-1} \ln(F/K) dK = K_{\min}^{\gamma} \ln(F/K_{\min})/\gamma + K_{\min}^{\gamma}/\gamma^2$, the boundary term at zero vanishing because $K^{\gamma} \ln(F/K) \rightarrow 0$. The upper-tail bounds follow from part (b) of the lemma. For $w(K) = K^{-2}$, $\int_{K_{\max}}^{\infty} K^{-1-\eta} dK = K_{\max}^{-\eta}/\eta$. For the logarithmic weight, the substitution $u = \ln(K/F)$ gives $F^{-\eta} \int_{\ell}^{\infty} |1-u| e^{-\eta u} du$, which is split at $u = 1$ and integrated by parts. $\square$

Remark 9. The bounds use only observed wing prices and an assumed tail exponent. A tail term whose weight has constant sign lies between 0 and its bound, taken with the sign of the weight; a term with a sign-changing weight lies in $[-B, B]$, where B bounds the integral of $|w|$. Because γ and η are not observed, risk-neutral variance and skewness are reported as intervals over a stated range of exponents, obtained by propagating the intervals of the individual contracts, never as point values. Between $K_{\min}$ and $K_{\max}$, P and C are taken from the calibrated smile and the integrals are to be computed by composite quadrature (research design, section 6), with F a node and each side integrated separately, because the out-of-the-money integrand has a kink at F where $P' - C' = 1$. With node spacing h on a side $[a, b]$ and an even number of subintervals for Simpson's rule, the composite trapezium error is at most $(b-a)h^2 \max |g''|/12$ if $g \in C^2[a, b]$, and the composite Simpson error at most $(b-a)h^4 \max |g^{(4)}|/180$ if $g \in C^4[a, b]$ (Süli and Mayers, 2003).

R5(b). Premium-adjusted call delta

The premium-adjusted delta, its non-monotonicity in the strike and the market practice of taking the strike to the right of the delta maximum are described by Reiswich and Wystup (2012). The proposition below makes the shape precise; the proof is mine.

Let $D \in (0, 1]$ and $h(K) = D(K/F)\Phi(d_{-}(K))$ for $K > 0$. With spot delta, D is the base-currency discount factor; with forward delta, $D = 1$.

Lemma 10. *The inverse Mills ratio $\lambda(d) = \varphi(d)/\Phi(d)$ is strictly decreasing on $\mathbb{R}$, with $\lambda(d) \rightarrow \infty$ as $d \rightarrow -\infty$ and $\lambda(d) \rightarrow 0$ as $d \rightarrow \infty$.*

Proof. Differentiation gives $\lambda'(d) = -\lambda(d)[d + \lambda(d)]$, so it suffices that $d\Phi(d) + \varphi(d) > 0$. This function has derivative $\Phi(d) > 0$ and tends to 0 as $d \rightarrow -\infty$; hence $d\Phi(d) + \varphi(d) = \int_{-\infty}^d \Phi(u) du > 0$. The limits follow from $\Phi \rightarrow 0$, $\varphi \rightarrow 0$ and the Mills-ratio asymptotics $\Phi(d) \sim \varphi(d)/|d|$ as $d \rightarrow -\infty$. Inequalities of this type for the Mills ratio are classical; see Sampford (1953). $\square$

Proposition 11. *The function h is continuous on $(0, \infty)$, tends to 0 as $K \downarrow 0$ and as $K \rightarrow \infty$, and satisfies $h < D$. There is a unique K^* such that h is strictly increasing on $(0, K^*]$ and strictly decreasing on $[K^*, \infty)$; $K^* = F \exp(-d^*s - s^2/2)$, where d^* is the unique solution of $\lambda(d) = s$. Each $\Delta \in (0, h(K^*))$ has exactly two preimages, one on each side of K^* .*

Proof. As $K \downarrow 0$, $d_{-} \rightarrow \infty$, so $h \rightarrow 0$. As $K \rightarrow \infty$, $d_{-} \rightarrow -\infty$ and, by the Mills-ratio asymptotics and the identity above, $K\Phi(d_{-}) \sim K\varphi(d_{-})/|d_{-}| = F\varphi(d_{+})/|d_{-}| \rightarrow 0$. Moreover $F\Phi(d_{+}) - K\Phi(d_{-})$ is the undiscounted call price, which is positive, so $h < D\Phi(d_{+}) < D$.

Since $d'_{-}(K) = -1/(Ks)$,

$$h'(K) = \frac{D}{F} \left[\Phi(d_{-}) - \frac{\varphi(d_{-})}{s} \right] = \frac{D\Phi(d_{-})}{Fs} [s - \lambda(d_{-})].$$

By the lemma, $s - \lambda(d)$ is strictly increasing in d and changes sign exactly once, at d^* . Because d_- is strictly decreasing in K , $h' > 0$ for $K < K^*$ and $h' < 0$ for $K > K^*$, where $d_-(K^*) = d^*$. The stated formula for K^* inverts d_- . The final claim follows from the intermediate value theorem and strict monotonicity on each side. $\square$

Remark 12. The market convention selects the preimage in $[K^*, \infty)$. At one month and 25Δ this root lies far to the right of K^* ; bracketing matters at long maturities and high volatilities.

R5(c). Delta-to-strike inversion with a smile

Proposition 13. *Let $\sigma : (0, \infty) \rightarrow (0, \infty)$ be continuously differentiable and let $\Delta_\phi(K) = \phi D \Phi(\phi d_+(K, \sigma(K)))$ be the pips spot delta (D the base-currency discount factor) or forward delta ($D = 1$) of a call ($\phi = 1$) or put ($\phi = -1$). On an interval J , if*

$$|\sigma'(K)| K \sqrt{\tau} |d_-(K, \sigma(K))| < 1 \quad \text{for all } K \in J,$$

then Δ_ϕ is strictly decreasing on J , and each delta attained on J is attained at exactly one strike in J .

Proof. With $d_+ = \ln(F/K)/(\sigma\sqrt{\tau}) + \sigma\sqrt{\tau}/2$, one has $\partial_K d_+ = -1/(K\sigma\sqrt{\tau})$ and $\partial_\sigma d_+ = -d_-/\sigma$. By the chain rule,

$$\frac{d}{dK} d_+(K, \sigma(K)) = -\frac{1}{K\sigma\sqrt{\tau}} \left[1 + K\sqrt{\tau} d_- \sigma'(K) \right],$$

which is negative on J under the stated condition. Since $\Delta'_\phi(K) = \phi^2 D \varphi(d_+) \frac{d}{dK} d_+ = D \varphi(d_+) \frac{d}{dK} d_+$, the function Δ_ϕ is strictly decreasing on J , and injectivity gives uniqueness. $\square$

Remark 14. In log-moneyness $k = \ln(K/F)$, $K\sigma'(K) = \partial\sigma/\partial k$, so the condition reads $|\partial\sigma/\partial k| \sqrt{\tau} |d_-| < 1$. Across the calibrated one-month smiles of the primary sample, its maximum over ± 4 ATM standard deviations is 0.76, so the smile delta-to-strike map is injective on that range at every month-end. For premium-adjusted delta $\phi D(K/F) \Phi(\phi d_-)$ the analogous derivative is $\phi(D/F) [\Phi(\phi d_-) + \phi K \varphi(d_-) \frac{d}{dK} d_-]$ with $\frac{d}{dK} d_- = -[1 + K\sqrt{\tau} d_+ \sigma'(K)]/(K\sigma\sqrt{\tau})$; numerically, the premium-adjusted put delta is strictly monotone and the call delta strictly decreasing to the right of its maximum on the same range at every month-end.

R6. Diffusive null of the hedge cost

The comparison of hedged and unhedged carry returns follows Jurek (2014), who reads the resulting ratio as an upper bound on the crash share and notes that an unlevered option hedge also gives up part of the diffusive premium (working-paper version of August 2013, Section 2.2). Farhi et al. (2015, Section 5.1, eqs. (8)–(9)) state the leading-order version of the result below: a put of delta Δ_P leaves a fraction $1 + \Delta_P$ of the Gaussian risk premium. The exact mean-value form and the error bound below are mine.

Let the undiscounted put value with lognormal forward G and volatility σ be

$$p(G) = K \Phi(-d_-(G)) - G \Phi(-d_+(G)), \quad d_\pm(G) = \frac{\ln(G/K) \pm s^2/2}{s}.$$

Using $K\varphi(d_-) = G\varphi(d_+)$, $p'(G) = -\Phi(-d_+(G))$.

Suppose that under $\mathbb{Q}$, S_τ is lognormal with mean F and log-variance s^2 , and that under $\mathbb{P}$ it is lognormal with the same log-variance and mean $Fe^{\lambda\tau}$ (no jumps, no volatility premium). A unit long forward costs nothing and has $\mathbb{P}$ -expected payoff $\mu_U = F(e^{\lambda\tau} - 1)$. Adding a put bought at its $\mathbb{Q}$ -value, with the premium financed to τ , gives $\mu_H = \mu_U + p(Fe^{\lambda\tau}) - p(F)$.

Proposition 15. *If $\lambda \neq 0$, the hedge-cost ratio $\theta_0 = 1 - \mu_H/\mu_U$ satisfies $\theta_0 = \Phi(-d_+(\xi))$ for some ξ between F and $Fe^{\lambda\tau}$, and*

$$|\theta_0 - \Phi(-d_1)| \leq \varphi(0) \frac{|\lambda|\sqrt{\tau}}{\sigma}, \quad d_1 = d_+(F).$$

Proof. By the mean value theorem, $p(Fe^{\lambda\tau}) - p(F) = -\Phi(-d_+(\xi)) F(e^{\lambda\tau} - 1)$ for some ξ between the two forwards, so $\theta_0 = -[p(Fe^{\lambda\tau}) - p(F)]/\mu_U = \Phi(-d_+(\xi))$. Since $|d_+(\xi) - d_+(F)| = |\ln(\xi/F)|/s \leq |\lambda|\tau/s$ and Φ is Lipschitz with constant $\varphi(0)$, the bound follows. $\square$

Remark 16. With no crash premium at all, hedging with a put of forward delta $-\Phi(-d_1)$ surrenders approximately that fraction of the carry premium: about 0.25 for a 25Δ put and 0.10 for a 10Δ put. This is the null against which the realised ratio θ_{UB} and the skew component of E3 are read.

R7. Entropy decomposition of currency premia

The relation between the exchange rate and the two pricing kernels is that of Backus, Foresi and Telmer (2001). Conditional entropy, its cumulant expansion and the entropy bound are as in Backus, Chernov and Zin (2014), who relate the bound to Bansal and Lehmann (1997) and Alvarez and Jermann (2005). The combination below, including the covered-interest-parity term, is derived here.

Fix t and write $\mathbb{E}_t$ for $\mathbb{E}[\cdot | \mathcal{F}_t]$. For a positive random variable M with $\mathbb{E}_t|\log M| < \infty$, the conditional entropy is $L_t(M) = \log \mathbb{E}_t M - \mathbb{E}_t \log M \geq 0$. Let $M, M^* > 0$ be $\mathcal{F}_{t+1}$ -measurable discount factors in L^2 that price domestic and foreign payoffs in their own currencies, with $\mathbb{E}_t M = e^{-r_t}$, $\mathbb{E}_t M^* = e^{-r_t^*}$, $\mathbb{E}_t|\log M|, \mathbb{E}_t|\log M^*| < \infty$ and $MS_{t+1}/S_t \in L^2$, where S is the domestic price of foreign currency. Write $m = \log M$, $\Delta s = \log(S_{t+1}/S_t)$, and $x_t = (f_t - s_t) - (r_t - r_t^*)$ for the covered-interest-parity basis. With exchange rates quoted in foreign currency per USD, as in Du, Tepper and Verdelhan (2018; NBER Working Paper 23170, eq. (4)), x_t is minus their basis.

Lemma 17. *Suppose every foreign payoff $X^* \in L^2(\mathcal{F}_{t+1})$ is traded and priced consistently in both currencies, $S_t \mathbb{E}_t[M^* X^*] = \mathbb{E}_t[MS_{t+1} X^*]$. Then $M^* = MS_{t+1}/S_t$ almost surely.*

Proof. The difference $D = M^* - MS_{t+1}/S_t$ lies in L^2 and satisfies $\mathbb{E}_t[DX^*] = 0$ for all $X^* \in L^2$. Taking $X^* = D$ gives $\mathbb{E}_t D^2 = 0$. $\square$

Proposition 18. *Under the lemma's conclusion, (i) $\mathbb{E}_t[\Delta s + r_t^* - r_t] = L_t(M) - L_t(M^*)$ and $\mathbb{E}_t[s_{t+1} - f_t] = L_t(M) - L_t(M^*) - x_t$. (ii) If $k(u) = \log \mathbb{E}_t e^{um}$ is finite near 0 and its Maclaurin series has radius of convergence greater than 1, then $L_t(M) = \sum_{j \geq 2} \kappa_{j,t}/j!$, where $\kappa_{j,t}$ are the conditional cumulants of m . (iii) For any gross return $R > 0$ with $\mathbb{E}_t[MR] = 1$ and $\mathbb{E}_t|\log R| < \infty$, $L_t(M) \geq \mathbb{E}_t \log R - r_t$.*

Proof. (i) By definition, $\mathbb{E}_t m = -r_t - L_t(M)$ and $\mathbb{E}_t m^* = -r_t^* - L_t(M^*)$. The lemma gives $\Delta s = m^* - m$, so $\mathbb{E}_t \Delta s = r_t - r_t^* + L_t(M) - L_t(M^*)$. The second identity follows from $s_{t+1} - f_t = \Delta s - (f_t - s_t) = \Delta s - (r_t - r_t^*) - x_t$. (ii) $L_t(M) = k(1) - k'(0)$ with $k'(0) = \kappa_{1,t}$, and $k(1) = \sum_{j \geq 1} \kappa_{j,t}/j!$ under the stated convergence. (iii) By Jensen's inequality, $0 = \log \mathbb{E}_t[MR] \geq \mathbb{E}_t m + \mathbb{E}_t \log R$, so $L_t(M) = -r_t - \mathbb{E}_t m \geq \mathbb{E}_t \log R - r_t$. $\square$

Remark 19. The identity in (i) needs only that some pair of discount factors prices bonds and exchange rates consistently; completeness makes the pair unique. Brandt, Cochrane and Santa-Clara (2006) show that the smoothness of exchange rates relative to the volatility of marginal utility implied by equity premia requires a very high degree of international risk sharing. Options on S reveal the risk-neutral law of $\Delta s = m^* - m$, not the separate cumulants of m and m^* , so they constrain but do not identify the split of $L_t(M) - L_t(M^*)$ across cumulant orders. Hedged and unhedged carry returns give two lower bounds on the same $L_t(M)$ through (iii).

R8. Local well-posedness of the smile calibration

Lemma 20. For $\beta = 1$ the Hagan, Kumar, Lesniewski and Woodward (2002) implied volatility $\sigma(K; \theta)$, $\theta = (\alpha, \rho, \nu)$, is infinitely differentiable on $\{K > 0\} \times (0, \infty) \times (-1, 1) \times [0, \infty)$.

Proof. For $\beta = 1$, $\sigma = \alpha(z/x(z)) [1 + (\rho\nu\alpha/4 + (2 - 3\rho^2)\nu^2/24)\tau]$ with $z = (\nu/\alpha) \ln(F/K)$ and $x(z) = \ln[(\sqrt{1 - 2\rho z + z^2} + z - \rho)/(1 - \rho)]$. Since $(1 - 2\rho z + z^2) - (z - \rho)^2 = 1 - \rho^2 > 0$, the square root exceeds $|z - \rho|$, so the argument of the logarithm is positive and x is smooth. Moreover $x'(z) = (1 - 2\rho z + z^2)^{-1/2} > 0$ and $x(0) = 0$, so $x(z, \rho) = z h(z, \rho)$ with $h(z, \rho) = \int_0^1 (1 - 2\rho t z + t^2 z^2)^{-1/2} dt$. Since $1 - 2\rho t z + t^2 z^2 \geq 1 - \rho^2 > 0$, the integrand is smooth and positive on $\mathbb{R} \times (-1, 1) \times [0, 1]$, so by differentiation under the integral h is smooth and positive, and $z/x(z) = 1/h(z, \rho)$ (equal to 1 at $z = 0$) is jointly smooth in (z, ρ) . Compositions and products of smooth maps are smooth. $\square$

Fix F, τ, D and the delta convention, and write $q = (\sigma_{ATM}, RR, BF)$ for the quotes. For parameters θ , let $K_A(\theta)$ solve $\ln(K/F) \mp \sigma(K; \theta)^2 \tau / 2 = 0$ (the delta-neutral-straddle strike, with the sign given by the premium convention). Let $K_c(\theta)$ and $K_p(\theta)$ solve $\Delta(K, \sigma(K; \theta)) = \pm 0.25$. Let $K_c^{ms}(q)$ and $K_p^{ms}(q)$ be the $\pm 25\Delta$ strikes at the single volatility $\sigma_{ATM} + BF$. Define $\Psi = (\Psi_1, \Psi_2, \Psi_3)$ by

$$\begin{aligned} \Psi_1 &= \sigma(K_A; \theta) - \sigma_{ATM}, & \Psi_2 &= \sigma(K_c; \theta) - \sigma(K_p; \theta) - RR, \\ \Psi_3 &= \sum_{i \in \{c, p\}} [V(K_i^{ms}, \sigma(K_i^{ms}; \theta)) - V(K_i^{ms}, \sigma_{ATM} + BF)], \end{aligned}$$

where V is the undiscounted Garman–Kohlhagen premium of the corresponding option. The calibrated smile solves $\Psi(\theta, q) = 0$. The market-strangle condition is that of Reiswich and Wystup (2012). For θ near θ^* , $K_A(\theta)$, $K_c(\theta)$ and $K_p(\theta)$ denote the C^1 branches through the strikes at θ^* given by the implicit function theorem under (i) and (ii) below; for a premium-adjusted call, the strike at θ^* is the root with $\partial_K \Delta < 0$ (the conventional branch), which is the root that `strike_from_delta_smile` selects.

Proposition 21. Let $\Psi(\theta^*, q^*) = 0$. Suppose that at (θ^*, q^*) : (i) $1 \mp K\sigma\sigma'(K)\tau \neq 0$ at K_A ; (ii) the strike derivative of $\Delta(K, \sigma(K; \theta^*))$ is non-zero at K_c and K_p (for pips delta, for instance under the condition of R5(c); for premium-adjusted delta the derivative in the remark after R5(c) must be checked directly, since a flat smile satisfies the R5(c) condition while the call derivative vanishes at K^*); (iii) for premium-adjusted delta, the 25Δ call strike at the flat volatility $\sigma_{ATM}^* + BF^*$ lies strictly to the right of the maximiser, $K_c^{ms} > K^*$ (equivalently $0.25 < h(K^*)$ in the notation of R5(b)); and (iv) $D_\theta \Psi(\theta^*, q^*)$ is non-singular. Then there are neighbourhoods $U \ni q^*$ and $W \ni \theta^*$ and a unique C^1 map $\theta : U \rightarrow W$ with $\Psi(\theta(q), q) = 0$. In particular θ^* is the only calibration in W with strikes on these branches, and the calibrated smile varies smoothly with the quotes.

Proof. Since x has the sign of z , $z/x(z) > 0$, so $\sigma(\cdot; \theta) > 0$ for all K exactly when $1 + (\rho\nu\alpha/4 + (2 - 3\rho^2)\nu^2/24)\tau > 0$. This holds at θ^* because $\sigma(K_A; \theta^*) = \sigma_{ATM} > 0$, hence near θ^* , and Δ and V are evaluated only at positive volatilities. By the lemma, each defining equation of K_A , K_c , K_p is then C^1 in (K, θ) . Condition (i) is the non-vanishing of the K -derivative of the first; condition (ii) that of the other two. The implicit function theorem therefore makes K_A , K_c and K_p C^1 functions of θ near θ^* . The strikes K_i^{ms} depend on q through $\sigma_{ATM} + BF$: in closed form for pips delta, $K_{c,p}^{ms} = F \exp(\frac{1}{2}s_q^2 \mp s_q \Phi^{-1}(0.25/D))$ with $s_q = (\sigma_{ATM} + BF)\sqrt{\tau}$; and by the implicit function theorem for premium-adjusted delta, since $h' < 0$ on (K^*, ∞) by the proof of R5(b) and condition (iii), and $\partial_K [-D(K/F)\Phi(-d_-)] = -(D/F)[\Phi(-d_-) + \varphi(d_-)/s] < 0$ for the put. Hence G is C^1 in (θ, q) near (θ^*, q^*) , and the implicit function theorem applied to G with (iv) gives the claim. $\square$

Remark 22. Condition (iv) is checked at every calibration through the finite-difference Jacobian J returned by the solver. J is taken in $x = (\ln \alpha, \operatorname{atanh} \rho, \ln \nu)$, with Ψ_3 divided by the flat-volatility strangle vega $\mathcal{V}$, so $J = \operatorname{diag}(1, 1, \mathcal{V}^{-1}) D_\theta \Psi \operatorname{diag}(\alpha, 1 - \rho^2, \nu)$, which is non-singular exactly when $D_\theta \Psi$ is. Across the primary sample $\operatorname{cond}(J)$ is at most 125 under the market-strangle reading and at most 634 under the smile-strangle reading. Under the smile-strangle reading, Ψ_3 is replaced by $\frac{1}{2}[\sigma(K_c; \theta) + \sigma(K_p; \theta)] - \sigma_{ATM} - BF$, and the proposition and its proof hold without condition (iii). The proposition is local: it does not assert existence of a solution for arbitrary quotes, and it does not exclude other solutions far from θ^* . The implementation therefore restarts from several points and records non-convergence.

The Garman–Kohlhagen PDE

Let $dS_t = (r_d - r_f)S_t dt + \sigma S_t dW_t$ under $\mathbb{Q}^d$ with constant coefficients, and let $c(t, S)$ be the Garman and Kohlhagen (1983) call price with strike K and expiry T ,

$$c(t, S) = S e^{-r_f(T-t)} \Phi(d_+) - K e^{-r_d(T-t)} \Phi(d_-), \quad d_\pm = \frac{\ln(S/K) + (r_d - r_f \pm \sigma^2/2)(T-t)}{\sigma \sqrt{T-t}}.$$

Proposition 23. (a) $c \in C^{1,2}([0, T] \times (0, \infty))$ satisfies

$$\partial_t c + (r_d - r_f)S \partial_S c + \frac{1}{2} \sigma^2 S^2 \partial_{SS} c - r_d c = 0,$$

$0 \leq c \leq S e^{-r_f(T-t)}$, and $c(t, S) \rightarrow (S - K)^+$ as $t \uparrow T$, locally uniformly in S . (b) If $u \in C^{1,2}([0, T] \times (0, \infty)) \cap C([0, T] \times (0, \infty))$ solves the same equation with $u(T, S) = (S - K)^+$ and $|u(t, S)| \leq a(1 + S^m)$ for some a, m , then $u = c$. No boundary condition at $S = 0$ is required.

Proof. (a) Write $\tau = T - t$. The identity $S e^{-r_f \tau} \varphi(d_+) = K e^{-r_d \tau} \varphi(d_-)$ gives $\partial_S c = e^{-r_f \tau} \Phi(d_+)$ and $\partial_{SS} c = e^{-r_f \tau} \varphi(d_+) / (S \sigma \sqrt{\tau})$. The same identity gives $\partial_t c = r_f S e^{-r_f \tau} \Phi(d_+) - r_d K e^{-r_d \tau} \Phi(d_-) - S e^{-r_f \tau} \varphi(d_+) \sigma / (2 \sqrt{\tau})$. Substitution verifies the equation. The bounds follow from $c = e^{-r_d \tau} \mathbb{E}[(S_T - K)^+ | S_t = S]$ and $\mathbb{E}[S_T | S_t = S] = S e^{(r_d - r_f)\tau}$. As $\tau \downarrow 0$, $\Phi(d_\pm) \rightarrow \mathbf{1}\{S > K\}$ for $S \neq K$ and $c(t, K) \rightarrow 0$, and c is Lipschitz in S with constant $e^{|r_f|T}$ uniformly in t , which gives locally uniform convergence.

(b) Fix (t, S) and let $\tau_n = \inf\{s \geq t : S_s \notin (1/n, n)\} \wedge (T - 1/n)$. Itô's formula for $e^{-r_d(s-t)} u(s, S_s)$ on $[t, \tau_n]$, the equation, and the boundedness of $\partial_S u$ on compact sets give $u(t, S) = \mathbb{E}[e^{-r_d(\tau_n-t)} u(\tau_n, S_{\tau_n})]$. Because S is a geometric Brownian motion started at $S > 0$, it reaches neither 0 nor ∞ before T , so $\tau_n \rightarrow T$ almost surely. Continuity of u on $[0, T] \times (0, \infty)$ then gives $u(\tau_n, S_{\tau_n}) \rightarrow (S_T - K)^+$. The growth bound and $\mathbb{E} \sup_{s \leq T} S_s^m < \infty$ allow dominated convergence, so $u(t, S) = e^{-r_d(T-t)} \mathbb{E}(S_T - K)^+ = c(t, S)$. The representation uses only values in the interior $(0, \infty)$, which the process never leaves, so no condition at $S = 0$ enters. Itô's formula and the Feynman–Kac representation are used in the forms given by Shreve (2004). $\square$

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